A Classification of Feedback Loops and Their Relation to Biases in Automated Decision-Making Systems

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EAAMO. Boston, November 01, 2023



Decision-Making Systems



Sequential Decision-Making Systems

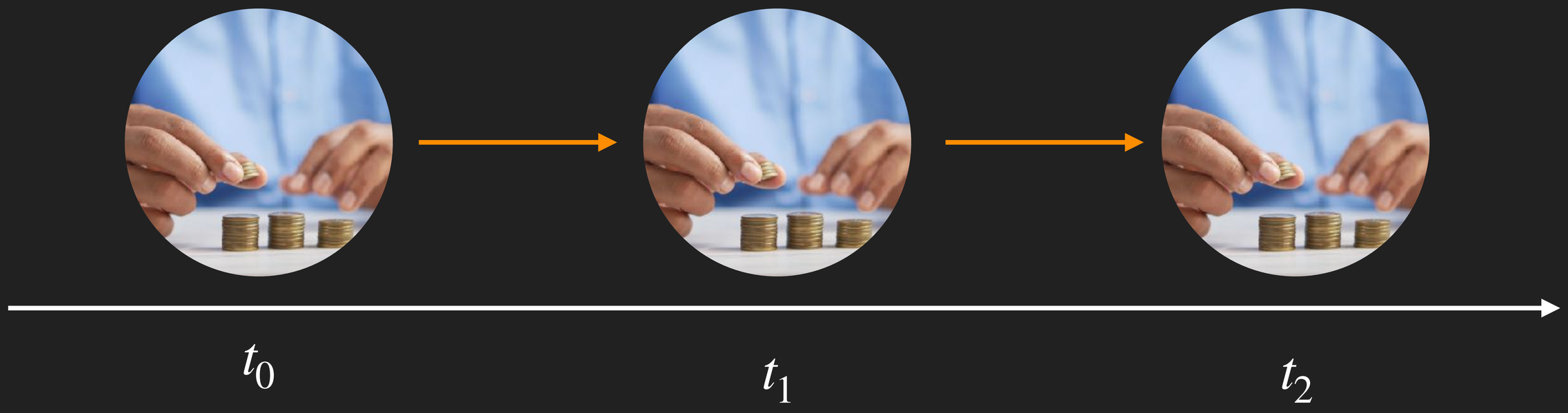






t_0 t_1 t_2

Sequential Decision-Making Systems



Sequential Decision-Making Systems with Feedback Loop



Sequential Decision-Making Systems with Feedback Loop



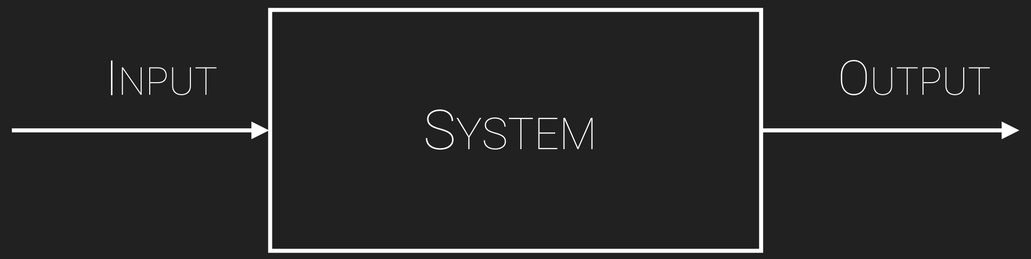
1. What is a FEEDBACK LOOP?

2. What TYPES of Feedback Loops occur in ML-based decision Making systems?

3. How do they affect BIASES?

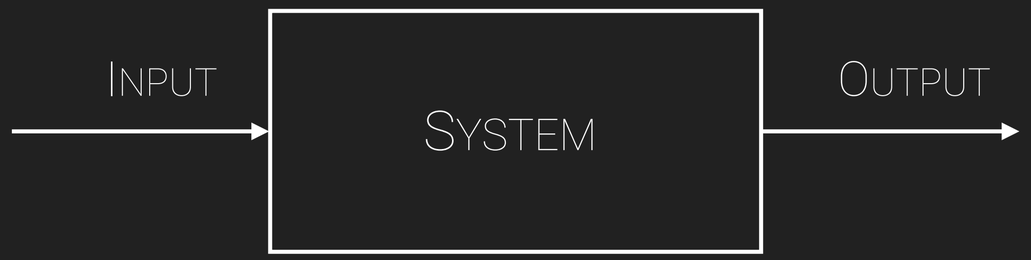
1. What is a Feedback Loop?

Dynamical Systems Theory

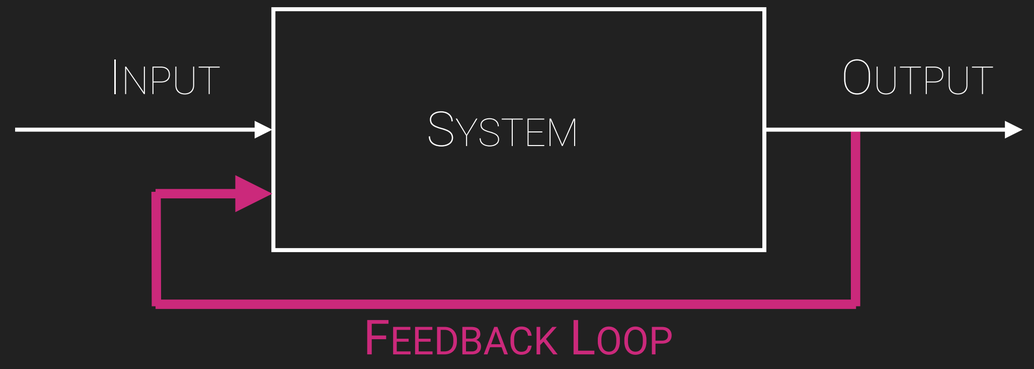


Open-loop system

Dynamical Systems Theory

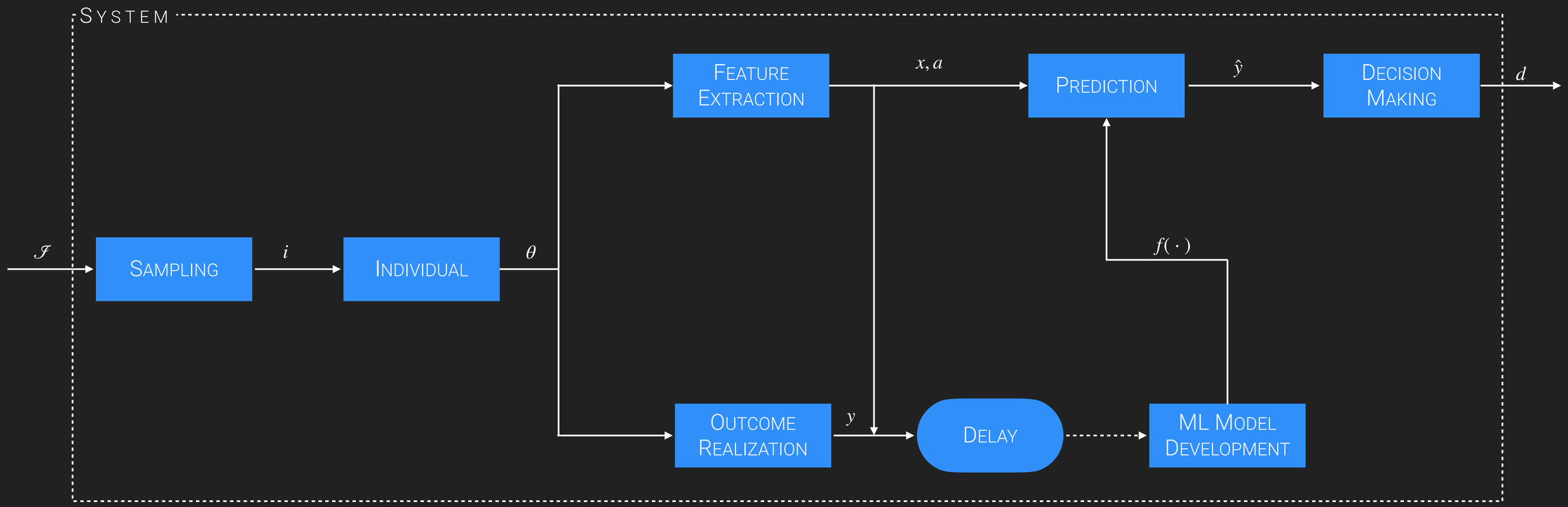


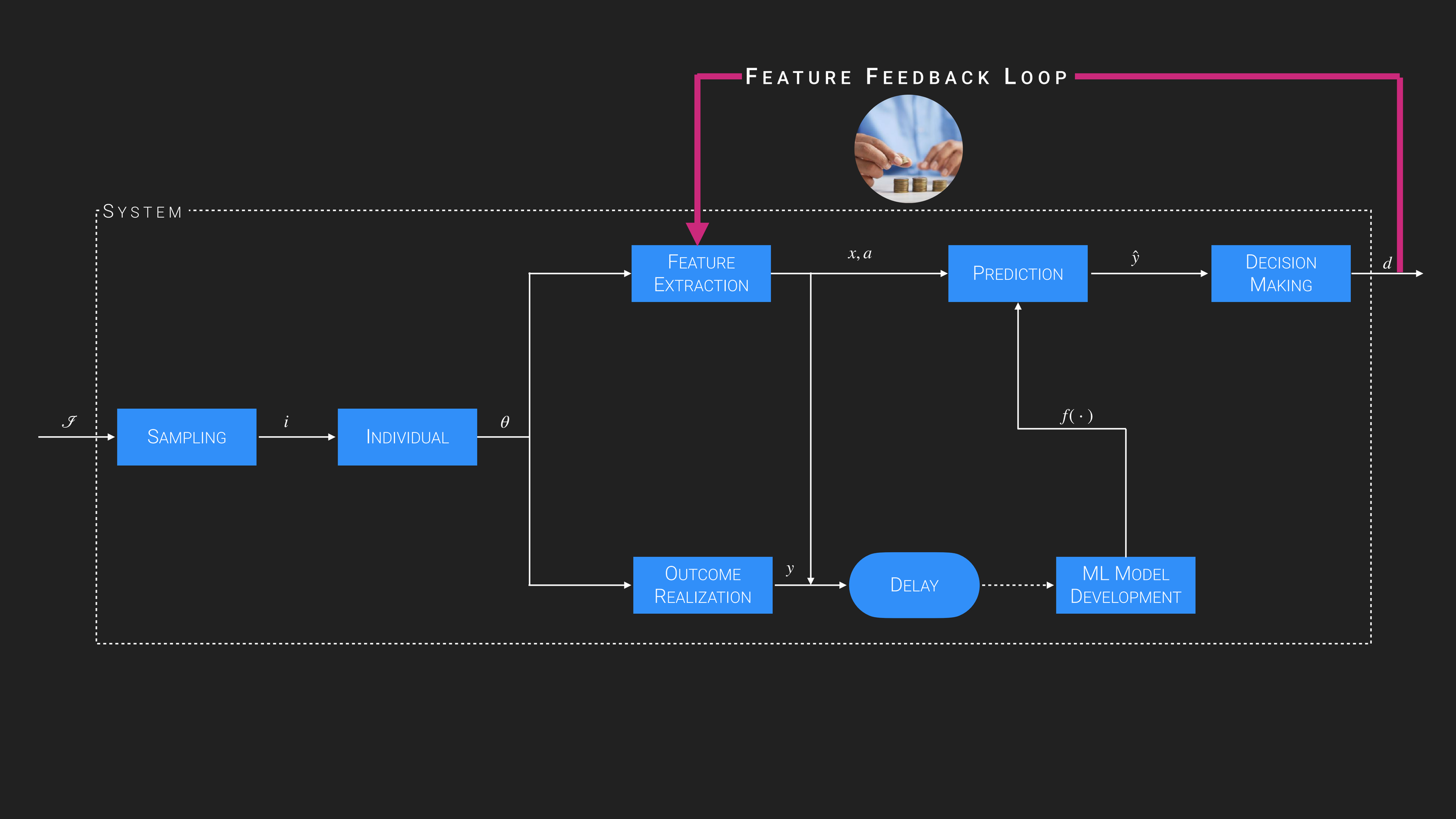
Open-loop system



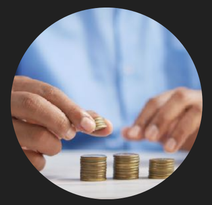
Closed-loop system

2. What **types** of Feedback Loops occur in ML Systems?

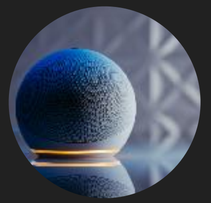

System
$\mathcal{I}$
Sampling
i
Individual
θ
Feature Extraction
x, a
Prediction
$\hat{y}$
Decision Making
d
Outcome Realization
y
Delay
ML Model Development
$f(\cdot)$


Feature Feedback Loop
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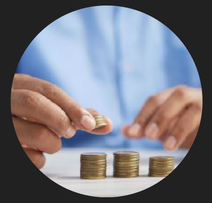




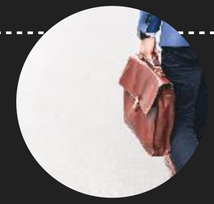
Individual Feedback Loop
Feature Feedback Loop
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Sampling Feedback Loop

Individual Feedback Loop

Feature Feedback Loop

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ML Model Development

ML model Feedback Loop

Outcome Feedback Loop

Case Study: Recommender Systems

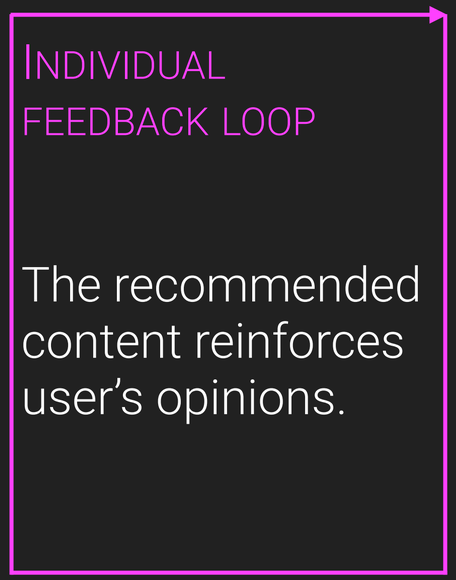
Prediction task:

Recommend *news to users by* predicting *their* clicking probability *based on individual features (e.g., click history).*

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Sampling feedback loop

Users that do not see content aligned with their view leave the platform.

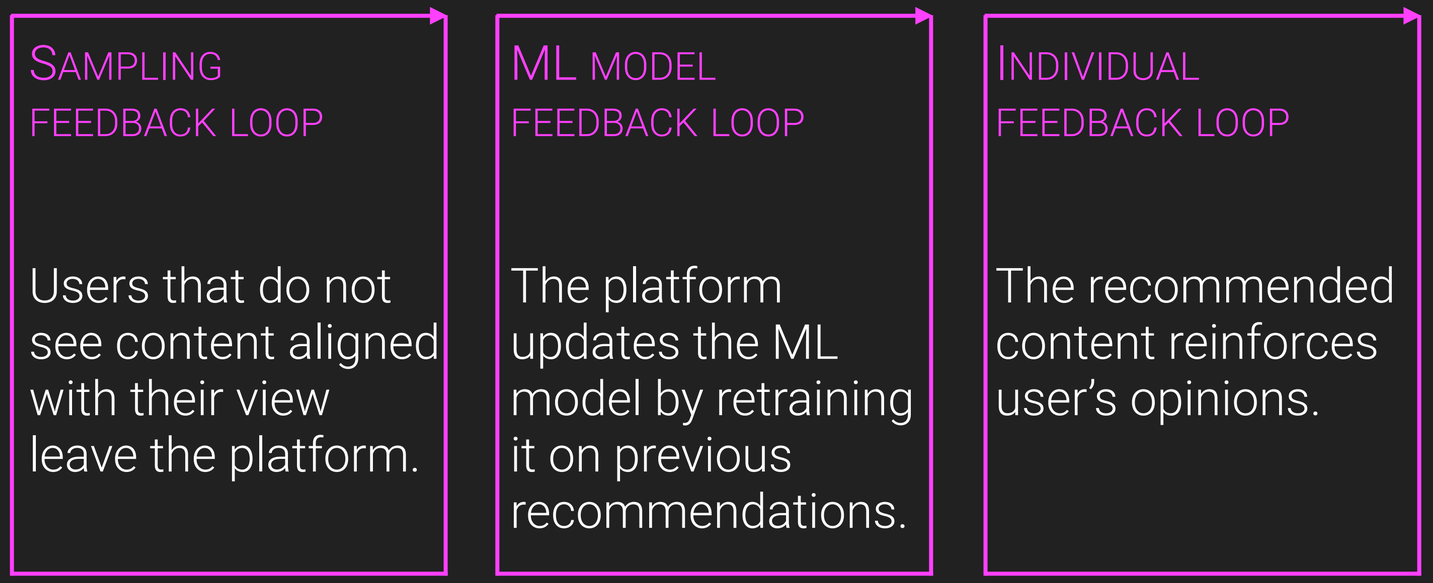
Individual feedback loop

The recommended content reinforces user's opinions.

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The platform updates the ML model by retraining it on previous recommendations.

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Feature feedback loop

The recommended content affects the features used for prediction.

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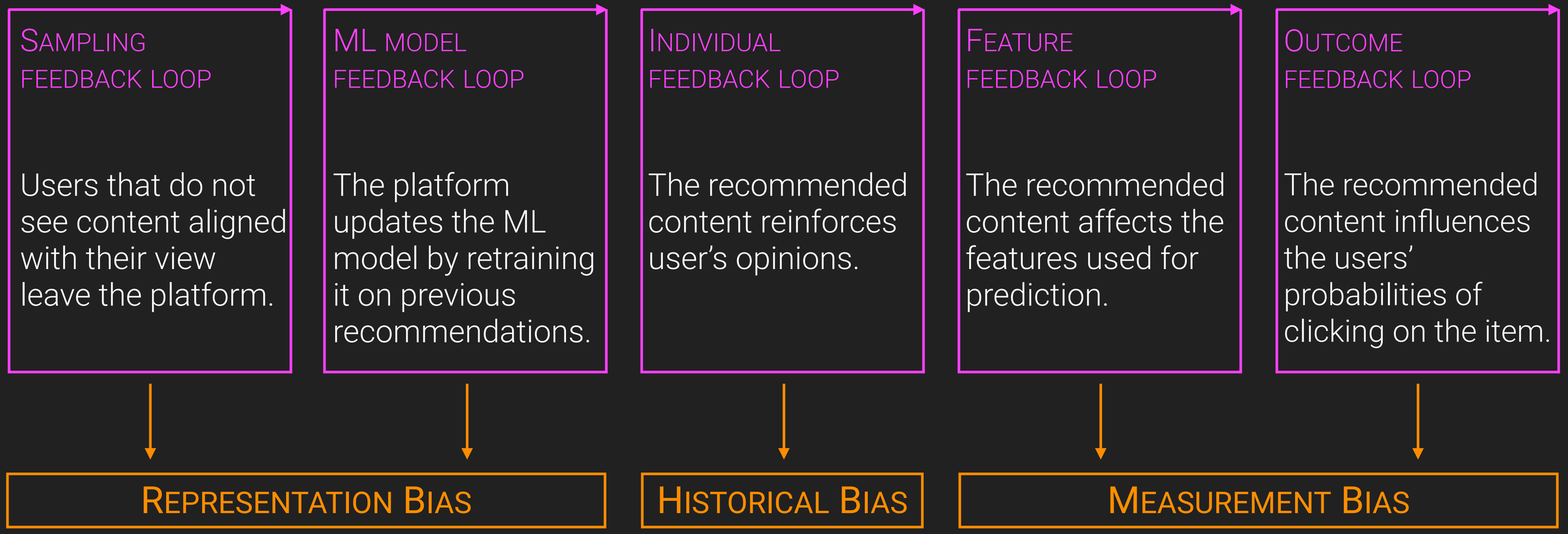
The recommended content influences the users' probabilities of clicking on the item.

3. How do Feedback Loops affect **BIASES**?

Case Study: Recommender Systems

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Takeaways

1. In ML the output decision feeds back into the system as an input, thus creates a Feedback Loop.

2. Feedback Loops come in different Shapes.

3. Different feedback loops can affect different Biases.

Outlook

1. Analyze, measure, and control Feedback Loops in ML systems.

2. Develop effective long-term unfairness mitigation techniques.



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FULL PAPER

